MILITARY SPECIFICATION SHEET

CABLES, RADIO FREQUENCY, FLEXIBLE, COAXIAL, DUAL, 125 OHMS, M17/16-RG23 (UNARMORED), M17/16-RG24 (ARMORED)

THIS CABLE USES PVC MATERIAL AND IS NOT TO BE USED IN AEROSPACE APPLICATIONS.

NOTE: THE AIR FORCE HAS RESTRICTED THE USE OF PVC IN AEROSPACE AND GROUND SUPPORT APPLICATIONS. CABLES WITH PVC JACKETING SHALL BE USED FOR RETROFIT PURPOSES ONLY UNTIL AN ALTERNATE JACKET IS APPROVED.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the cables described herein shall consist of this specification and the latest issue of MIL-C-17.

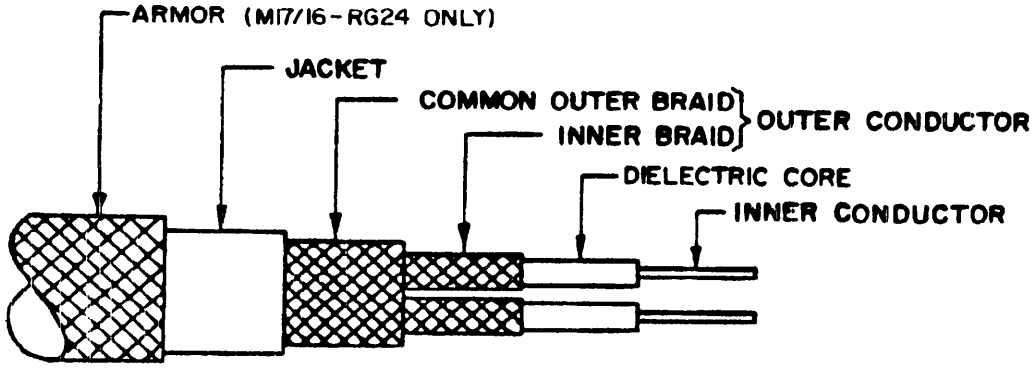


FIGURE 1. Configuration.

Ⓓ denotes changes

FSC 6145

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TABLE I. Description.

Component	Construction details
Inner conductors	Two conductors. Seven strands of bare copper wire, each strand 0.0285 inch each diameter. One strand of one conductor shall be tinned copper for identification. Diameter of each conductor: .0855 ±.0010.
Dielectric cores	Two cores, laid parallel to each other. Type A-1: Solid polyethlyene, each core. Diameter of each core: .380 inch ±.014.
Outer conductor:	Braid of AWG No.34, bare, copper wire. Two individual inner braids laid parallel to each other, under a common outer braid.
Inner braids (Individual)	Diameter of each braid: 0.407 inch maximum. Coverage : 94.0% nominal Carriers : 24 Ends : 9 Picks/inch: 10.5 ±10%
Outer braid (Common)	Diameter of major axis: .826 inch maximum. Coverage : 96.6% nominal Carriers : 24 Ends : 9 Picks/inch: 12.5 ±10%
Jacket	Type IIa: PVC. Diameter: 0.650 inch ±.020 by .945 inch ±.020.
Armor (M17/16-RG24 only)	Single braid of aluminum-alloy wire. Diameter: .735 by 1.034 maximum inches.

ENGINEERING INFORMATION:

Continuous working voltage: 5,000 V rms, maximum.

Operating frequency: 1 GHz, maximum.

Velocity of propagation: 65.9 percent, nominal.

Operating temperature range: -40°C to +85°C.

Inner conductor properties:

(D) DC resistance (maximum at 20°C): .191 ohm per 100 feet (each conductor).

Elongation: 25 percent, minimum.

Tensile strength: Not applicable.

Engineering note: This cable is useful in "balanced cables" low temperature applications (see connector per MIL-C-3655).

supersession data: See table II.

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REQUIREMENTS:

Dimensions, configuration, and description: See figure 1 and table I.

Environmental and mechanical:

(D) Visual and mechanical examination:

Out-of-roundness: Not applicable.

Eccentricity: Not applicable.

Adhesion of conductors:

(D) Inner conductor to core: 9 pounds, minimum; 90 pounds, maximum.

Aging stability: +98°C ±2°C.

Stress crack resistance: Not applicable.

Outer conductor integrity: Not applicable.

(D) Cold bend: -40°C ±2°C.

Dimensional stability: +85°C ±2°C.

Inner conductor from core: .200 inch, maximum.

Inner conductor from jacket: .400 inch, maximum.

Contamination: Applicable.

Bendability: Not applicable.

Flammability: Not applicable.

Weight:

M17/16-RG23: 53 pounds per 100 foot, maximum.
M17/16-RG24: 73 pounds per 100 foot, maximum.

Electrical:

Continuity: Applicable.

(D) Spark test: 8,000 V rms, +10%, -0%.

(D) Voltage withstanding: 12,000 V rms, +10%, -0%.

Insulation resistance: Not applicable.

Corona extinction voltage: 7,000 V rms, minimum.

Characteristic impedance: 125 ohms ±5.

Attenuation: 5.2 dB per 100 feet, maximum, at 400 MHz (for each individual coaxial core).

Structural return loss: Not applicable.

Capacitance: 13 pF per foot, maximum.

(D) Capacitance unbalance: 5 percent maximum.

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Ⓓ Transmission unbalance: 12 percent, maximum, over the 100 to 160 MHz frequency band.

Mechanically induced noise voltage: Not applicable.

Time delay: Not applicable.

Part number: See table II.

Supersession data: See table II.

TABLE II. Cross-reference of part number.

Part number	Superseded part number or type designation
M17/16-RG23	RG23A
M17/16-RG24	RG24A per MIL-C-17/17

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - SH, TD
Air Force - 11, 17, 99
DLA - ES, IS

User activities:
Army - AR, AT, ME
Navy - AS, MC, OS
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Army - CR

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